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View Abstract

Project: AI-Driven Solar-Biogas System for Sustainable Communities | Team: Espada
Track: Innovation Challenge | Theme: SDG 11: Sustainable Cities and Communities
Status: **Accepted** | Your Role: **Team leader**

Project Information

Project Name: AI-Driven Solar-Biogas System for Sustainable Communities

Team Name: Espada

Track: **Innovation Challenge**

Theme: **SDG 11: Sustainable Cities and Communities**

Referral Code: PA069

Team Leader

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Team Leader

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Year & Department: Associate Professor
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State: Tamil Nadu

Country: India

Gender: Female

IEEE Membership: 98059247

Uploads & Supporting Material

Presentation Slides

yesist ppt final .pdf

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Project Report

BIO GAS SYSTEM (1).pdf

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Problem Statement

Describe the specific problem statement your project addresses:

The project addresses the critical challenges of inefficient organic waste management, increasing cooking fuel costs, and growing dependence on fossil fuels in rural and semi-urban communities. A significant amount of household kitchen waste is generated daily, but most of it is either dumped in landfills or improperly managed, leading to environmental pollution and greenhouse gas emissions. At the same time, households rely heavily on LPG cylinders and conventional fuels, which are becoming increasingly expensive and contribute to carbon emissions. Although biogas technology exists, many systems lack proper monitoring, efficiency optimization, safe storage mechanisms, and structured distribution models. Additionally, there is no integrated digital platform to manage gas ordering, waste contribution tracking, and fair cost distribution among users. This limits scalability and reduces user adoption. The proposed Smart Solar–Biogas Integrated Community Energy System solves these issues by converting household organic waste into biogas, optimizing production using solar-powered sensors and AI monitoring, storing gas safely in cylinder-like units, and enabling app-based ordering and distribution. The system links waste contribution to fuel cost reduction, creating both





sensors and AI monitoring, storing gas safely in cylinder-like units, and enabling app-based ordering and distribution. The system links waste contribution to fuel cost reduction, creating both environmental and economic benefits. This project aims to establish a sustainable, technology-driven, community-level clean energy solution that reduces waste, lowers fuel expenses, and minimizes environmental impact.

Explain why this problem is important in the current global/local context:

The problem of improper organic waste management and rising dependence on fossil fuels is highly relevant in today's global and local context. Globally, organic waste dumped in landfills produces methane, a powerful greenhouse gas that contributes significantly to climate change. At the same time, continued reliance on LPG and other fossil fuels increases carbon emissions and accelerates environmental degradation. Locally, many households face rising cooking fuel prices and inconsistent energy access, especially in rural and semi-urban areas. Poor waste disposal practices also create sanitation issues, foul odors, and health risks. Rapid population growth and urbanization have further increased the volume of daily waste, while sustainable waste-to-energy solutions remain limited. Energy security is another growing concern, as fuel price fluctuations and supply disruptions directly affect household expenses.

Who are the primary beneficiaries or target users? (communities, industries, etc.):

Rural and Semi-Urban Households – Families who require affordable and clean cooking fuel.

Community Kitchens – Schools, hostels, temples,



access, especially in rural and semi-urban areas. Poor waste disposal practices also create sanitation issues, foul odors, and health risks. Rapid population growth and urbanization have further increased the volume of daily waste, while sustainable waste-to-energy solutions remain limited. Energy security is another growing concern, as fuel price fluctuations and supply disruptions directly affect household expenses.

Who are the primary beneficiaries or target users? (communities, industries, etc.):

Rural and Semi-Urban Households – Families who require affordable and clean cooking fuel.

Community Kitchens – Schools, hostels, temples, and local food centers that need a reliable cooking gas supply.

Farmer Groups and Dairy Farms – Contributors of agricultural and organic waste for biogas production.

Local Self-Governments / Panchayats – Authorities managing waste, sanitation, and community energy systems.

Small-Scale Entrepreneurs / Energy Cooperatives – Individuals or groups involved in gas storage, cylinder distribution, and app-based management services.

Innovation & Originality

What is your unique approach to solving this problem?:

Our unique approach integrates renewable energy, smart automation, structured gas distribution, and digital connectivity into a single, community-driven





Innovation & Originality

What is your unique approach to solving this problem?:

Our unique approach integrates renewable energy, smart automation, structured gas distribution, and digital connectivity into a single, community-driven ecosystem. Unlike conventional biogas plants that operate with minimal monitoring and limited scalability, our system combines solar-powered sensor networks and AI-based optimization to ensure maximum methane production and safe operation. Real-time data from temperature, pH, and gas pressure sensors is analyzed using intelligent algorithms to automatically regulate stirring, heating, and gas flow. This dynamic optimization significantly improves efficiency and reliability compared to traditional digesters. Solar photovoltaic energy powers the monitoring and control units, ensuring uninterrupted operation even in areas with limited grid access.

Additionally, the system introduces a structured model for compressing and storing surplus biogas in cylinders, enabling organized distribution and revenue generation. A dedicated mobile application allows users to monitor gas availability, track efficiency, book refills, and view carbon savings, creating transparency and accountability. By linking waste contribution to fuel cost reduction and combining clean energy with digital management, our approach establishes a scalable, data-driven, and economically sustainable community energy model aligned with global climate and sustainability goals.

How does your solution differ from existing solutions or competitors?



How does your solution differ from existing solutions or competitors?:

Existing biogas solutions primarily focus on waste-to-energy conversion but often lack intelligent monitoring, structured distribution models, and digital integration. Most traditional biogas plants operate manually, resulting in inconsistent methane production, low efficiency, and limited scalability. Additionally, many systems depend on grid electricity and do not provide mechanisms for transparent gas tracking or surplus revenue generation. Our solution differs by integrating solar-powered smart monitoring, AI-driven optimization, structured gas compression and storage, and an app-based distribution platform into a unified ecosystem. Real-time sensor data enables dynamic control of digestion parameters, significantly improving gas yield and operational reliability. Solar energy ensures uninterrupted monitoring even in off-grid or power-deficient regions. Furthermore, unlike conventional systems that supply gas only locally without structured management, our model allows surplus biogas to be compressed into cylinders for organized distribution and community-level revenue generation. The dedicated mobile application enhances transparency by enabling gas booking, efficiency tracking, and carbon emission reporting. By combining renewable energy, artificial intelligence, circular waste management, and digital accessibility, our solution transforms traditional biogas systems into a scalable, technology-driven clean energy platform suited for modern sustainability challenges.

Literature Survey:

- **Journal name:** Designing a Novel Explainable Artificial Intelligence Framework for Biogas



modern sustainability challenges.

Literature Survey:

- **Journal name:** Designing a Novel Explainable Artificial Intelligence Framework for Biogas Generation From Organic Waste
Justification: The proposed research addresses the instability and inefficiency of micro-scale anaerobic digestion (AD) systems caused by fluctuating feedstock and environmental conditions. Traditional and black-box AI models lack both predictive reliability and interpretability, limiting real-time operational decision-making. The explainable XCO-Net framework, enhanced with deep temporal learning and the Starfish Optimization Algorithm, improves forecasting accuracy and yield consistency. This approach transforms micro-AD systems into intelligent, adaptive, and sustainable waste-to-energy platforms aligned with low-carbon energy goals.
- **Journal name:** Development of an AI-Powered Biogas Production Prediction Model for Smart Gobar Gas Systems
Justification: This paper demonstrates the practical use of AI models for predicting biogas production in small-scale systems. It supports the feasibility of integrating intelligent forecasting into decentralized AD units. The study justifies the need for improved prediction accuracy and system automation in micro-biogas applications.
- **Journal name:** Biogas production management systems with model predictive control of anaerobic digestion processes
Justification: This research highlights the importance of advanced control strategies for





improved prediction accuracy and system automation in micro-biogas applications.

- **Journal name:** Biogas production management systems with model predictive control of anaerobic digestion processes
Justification: This research highlights the importance of advanced control strategies for stabilizing anaerobic digestion performance. It validates the necessity of predictive and adaptive frameworks to manage fluctuating process parameters. The work supports the integration of intelligent optimization in biogas systems.
- **Journal name:** Feature Engineering and Supervised Machine Learning to Forecast Biogas Production during Municipal Anaerobic Co-Digestion
Justification: The paper emphasizes the role of machine learning and feature engineering in improving forecasting accuracy. It demonstrates how data-driven models outperform traditional empirical approaches. This strengthens the justification for adopting deep temporal learning in biogas prediction systems.
- **Journal name:** Artificial Intelligence-Based Modeling of Biogas Production Using ANN and ANFIS
Justification: This study compares AI techniques such as ANN and ANFIS for biogas yield estimation. It highlights the nonlinear and time-dependent nature of AD systems, supporting the use of advanced neural architectures. However, it also reveals limitations in interpretability, reinforcing the need for explainable AI frameworks.





Have you implemented any novel technologies or frameworks (e.g., AI, IoT, ML, Blockchain)?:

Yes

What makes your solution original (e.g., novel concept, data-driven design, research-backed idea)?:

Our solution is original because it integrates explainable artificial intelligence, deep temporal learning, and biologically inspired optimization into a unified framework specifically designed for micro-scale anaerobic digestion (AD) systems. While existing biogas models primarily focus on prediction accuracy, they often function as black-box systems without interpretability. In contrast, our proposed Explainable Conjoined O-Net (XCO-Net) not only forecasts biogas production but also provides interpretable outputs that help operators understand the influence of feedstock composition, environmental variables, and loading patterns on system performance. Another key novelty lies in the integration of the Starfish Optimization Algorithm for adaptive hyperparameter tuning. Unlike conventional grid or random search methods, this biologically inspired strategy enhances model efficiency by dynamically navigating complex parameter spaces. This improves prediction stability and yield consistency under fluctuating real-world conditions. Furthermore, the framework is specifically tailored for decentralized, residential micro-AD units rather than large industrial plants, addressing an underexplored segment in waste-to-energy research. By combining data-driven intelligence, explain ability , adaptive optimization, and real-time applicability, the solution transforms traditional biogas systems into cognitive, low-carbon, and scalable energy platforms aligned





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Implementation Details

Describe the technical architecture / workflow of your project:

Organic waste is collected and fed into the micro-scale anaerobic digestion (AD) unit.

IoT sensors measure key parameters such as temperature, pH, loading rate, and gas output in real time.

Sensor data is transmitted to a cloud database for secure storage and processing.

The data undergoes preprocessing, including cleaning, normalization, and feature engineering. Processed time-series data is fed into the XCO-Net deep temporal learning model.

The Starfish Optimization Algorithm tunes model hyperparameters for optimal accuracy and stability.

The model generates real-time biogas yield predictions with interpretable insights.

Results are displayed on a dashboard or mobile app for operator monitoring and decision support.





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Results are displayed on a dashboard or mobile app for operator monitoring and decision support. The system continuously updates and retraines using new incoming data to improve performance over time.

What technologies, platforms, or programming languages did you use? (e.g., Python, React, Arduino, etc.):

Python, JavaScript ,Dart, SQL, HTML, CSS

What stage is your project currently in?:

Prototype

What challenges did you face during development and how did you overcome them?:

During development, we faced several technical and practical challenges. One major challenge was handling the nonlinear and time-dependent behavior of anaerobic digestion (AD) systems. Biogas production fluctuates due to variations in feedstock composition, temperature, and loading rates. We addressed this by implementing deep temporal learning models capable of capturing sequential patterns and long-term dependencies in the data. Another challenge was hyperparameter tuning for achieving stable and accurate predictions. Manual tuning was inefficient and inconsistent, so we integrated the Starfish Optimization Algorithm to automate adaptive parameter selection, significantly improving model performance and convergence stability .Data quality and availability also posed difficulties, as micro-scale AD datasets are often limited and noisy. We overcame this through rigorous preprocessing, normalization, feature engineering, and validation techniques to enhance model robustness. Finally, ensuring model





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Milestones & Metrics





Milestones & Metrics

List your key milestones achieved so far (e.g., prototype completed, pilot run, user feedback collected):

Completed conceptual design of the Explainable AI-based biogas forecasting framework (XCO-Net)
Developed and tested the deep temporal learning prototype using real-world micro-AD dataset.

Successfully implemented the Starfish Optimization Algorithm for automated hyperparameter tuning.

Achieved strong model performance (RMSE = 0.38 m³/day, NNSE = 0.92) with 39% improvement over baseline methods.

Designed the system architecture including IoT data integration and cloud-based workflow.

Developed initial dashboard/app prototype for real-time monitoring and prediction visualization.

Conducted preliminary validation and comparative analysis with existing machine learning models.

Provide data or metrics showing your project's performance or validation results.:

The proposed XCO-Net model was validated using real-world micro-scale anaerobic digestion data under varying operational conditions.

It achieved a Root Mean Square Error (RMSE) of 0.38 m³/day and a Normalized Nash–Sutcliffe Efficiency (NNSE) of 0.92, indicating high prediction accuracy.

The system showed a 39% improvement in prediction consistency compared to baseline ANN and regression models.

The Starfish Optimization Algorithm enhanced training stability and reduced overfitting.

Cross-validation confirmed robust performance





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The Starfish Optimization Algorithm enhanced training stability and reduced overfitting.

Cross-validation confirmed robust performance and reliable real-time biogas forecasting.

How do you measure success or progress? (KPIs, quantitative or qualitative metrics):

Quantitative KPIs include prediction accuracy metrics such as RMSE, NNSE, and percentage improvement over baseline models. We also track biogas yield consistency, reduction in forecasting error, model convergence time, and system stability under varying operational conditions.

Additional technical indicators include hyperparameter optimization efficiency and cross-validation performance scores.

Operational KPIs include improvement in gas production stability, reduction in downtime, and enhanced decision-making accuracy for feedstock management.

Qualitative metrics involve interpretability clarity of model outputs, ease of integration with micro-AD systems, and user/operator feedback on dashboard usability and real-time monitoring support.





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What are your short-term goals (next 6–12 months)?:

Improve and test the XCO-Net model with more real-world data.

Deploy a pilot system in a small residential or community biogas unit.

Enhance the explainable AI dashboard for clearer real-time insights.

Optimize the system for cloud deployment and scalability.

Publish the research and explore partnerships for field implementation.

What are your long-term milestones (1–3 years)?:

Deploy the system in multiple community and rural micro-AD units for large-scale validation.

Improve the AI framework with advanced adaptive learning and real-time control integration.

Develop a fully automated smart biogas management platform with IoT and mobile app support.

Achieve commercial readiness through industry partnerships and pilot commercialization.

Contribute to low-carbon energy adoption by expanding the solution to regional and national levels.



Sustainability & Scalability

How does your project contribute to environmental, social, and/or economic



Sustainability & Scalability

How does your project contribute to environmental, social, and/or economic sustainability?:

Our project contributes significantly to environmental, social, and economic sustainability by transforming traditional anaerobic digestion (AD) systems into intelligent, efficient, and scalable waste-to-energy platforms. From an environmental standpoint, the framework improves biogas yield stability and reduces methane emissions caused by unmanaged organic waste. By optimizing digestion parameters using explainable AI and adaptive learning, the system enhances renewable energy generation while decreasing reliance on fossil fuels. This directly supports carbon reduction goals and promotes a circular economy by converting organic waste into clean energy. Socially, the project strengthens decentralized and community-level energy access. Reliable biogas forecasting reduces uncertainty in fuel availability, especially in rural and off-grid regions. The explainable nature of the AI model ensures transparency, allowing operators and stakeholders to understand system behavior and make informed decisions. This builds trust, improves technical awareness, and encourages broader adoption of renewable energy technologies. Economically, improved forecasting accuracy reduces operational inefficiencies, feedstock wastage, and maintenance costs. Stable gas production enhances productivity and long-term profitability of micro-AD systems. The scalable architecture also opens opportunities for commercialization, green entrepreneurship, and





accuracy reduces operational inefficiencies, feedstock wastage, and maintenance costs. Stable gas production enhances productivity and long-term profitability of micro-AD systems. The scalable architecture also opens opportunities for commercialization, green entrepreneurship, and local employment generation. Overall, the project integrates advanced artificial intelligence with sustainable waste management principles, creating a resilient, low-carbon, and socially inclusive energy solution aligned with global clean energy and climate objectives.

What strategies ensure your project's long-term viability (financial, operational, or community-based)?:

The system reduces operational costs by improving biogas yield stability and minimizing downtime.

A scalable subscription or service-based model supports long-term financial sustainability. Cloud-based architecture enables remote monitoring, updates, and continuous model improvement.

Explainable AI ensures transparency, increasing operator trust and ease of maintenance.

Community engagement and training programs promote local adoption and energy independence. Continuous data-driven optimization ensures long-term operational reliability and growth.

How can your project be scaled or replicated in other regions or contexts?:

The project uses a modular, software-driven AI framework that can be integrated with existing anaerobic digestion systems. The model can be retrained with region-specific data to adapt to different climates, feedstock types, and waste





How can your project be scaled or replicated in other regions or contexts?:

The project uses a modular, software-driven AI framework that can be integrated with existing anaerobic digestion systems .The model can be retrained with region-specific data to adapt to different climates, feedstock types, and waste patterns. Cloud-based deployment allows multiple biogas units to be monitored and managed centrally. Standard IoT sensor integration ensures easy replication across rural, urban, or agricultural settings. The dashboard can be localized into different languages for broader accessibility. A scalable SaaS model enables expansion through partnerships with governments, NGOs, and clean energy startups.

What partnerships or collaborations could help expand your impact?:

Collaboration with local governments and municipal bodies to integrate the system into community waste-to-energy programs.

Partnerships with renewable energy startups and biogas plant manufacturers for large-scale deployment and commercialization.

Collaboration with universities and research institutions for continuous model improvement and validation studies.

Engagement with NGOs working in rural development and sustainability to implement pilot projects in underserved communities.

Partnerships with IoT and cloud service providers to enhance real-time monitoring and scalable deployment.

Support from clean energy incubators and climate-tech accelerators to scale funding, mentorship, and global outreach.





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Cost & Resource Optimization

Provide an overview of your budget breakdown (materials, tools, labor, etc.):

IoT Sensors & Hardware (₹25,000 – ₹40,000)

Anaerobic Digestion Setup / Prototype Unit (₹30,000 – ₹60,000)

Software & Cloud Services (₹10,000 – ₹25,000 per year)

Development Tools & Testing (₹10,000 – ₹20,000)

Labor & Technical Development (₹50,000 – ₹1,00,000)

Contingency & Miscellaneous (₹15,000 – ₹25,000)

Approximately ₹1,40,000 – ₹2,70,000

Describe how you plan to optimize or minimize costs without reducing impact.:

Utilize open-source software frameworks and libraries to eliminate licensing costs while maintaining high-performance AI development.

Deploy cost-effective IoT hardware and locally





Describe how you plan to optimize or minimize costs without reducing impact.:

Utilize open-source software frameworks and libraries to eliminate licensing costs while maintaining high-performance AI development. Deploy cost-effective IoT hardware and locally sourced materials to reduce procurement and maintenance expenses.

Adopt a modular and scalable architecture, allowing incremental upgrades instead of complete system overhauls.

Implement cloud-based infrastructure with optimized resource allocation to minimize computing and storage costs.

Partner with academic institutions, NGOs, and local communities to share infrastructure, expertise, and pilot-testing facilities.

Continuously refine the AI model to improve computational efficiency, reducing long-term operational and energy costs without compromising prediction accuracy.

Have you received any funding, sponsorship, or institutional support?:

No

What additional resources or support would you need to scale further?:

Seed Funding & Grants

Government & Policy Support

Industry Partnerships

Technical Infrastructure

Field Deployment & Community Networks

Research & Academic Collaboration





Community & Impact

Who benefits directly from your project, and how? (communities, regions, industries, etc.):

Rural and semi-urban communities ,Micro-scale biogas plant operators, Agricultural farms and livestock owners, Municipal waste management authorities,
Renewable energy startups and clean-tech industries, Environmental stakeholders and policymakers

Have you engaged with any stakeholders or local communities in your initiative?:

No

Provide quantifiable or anecdotal evidence of social/environmental impact (data, testimonials, etc.):

The system improved biogas prediction consistency by 39%, with high accuracy (RMSE = 0.38 m³/day, NNSE = 0.92).,Stable forecasting can increase usable biogas output by an estimated 15–25% compared to traditional methods. Improved waste utilization reduces methane emissions from unmanaged organic waste.

Communities benefit from lower LPG usage and reduced monthly energy expenses. Real-time monitoring minimizes downtime and improves operator confidence and system reliability.

What are your plans for monitoring and evaluating ongoing impact?:

Continuously track model performance using KPIs such as RMSE, NNSE, and prediction consistency. Monitor biogas yield stability and compare actual





What are your plans for monitoring and evaluating ongoing impact?:

Continuously track model performance using KPIs such as RMSE, NNSE, and prediction consistency. Monitor biogas yield stability and compare actual vs. predicted output regularly.

Measure reduction in feedstock waste, downtime, and operational costs over time.

Assess environmental impact through estimated methane emission reduction and fossil fuel displacement.

Collect user and operator feedback to evaluate usability and decision-support effectiveness.

Perform periodic system audits and retraining of the AI model to maintain long-term accuracy and reliability.

Future Roadmap & Vision

What are your next steps after this submission (implementation, expansion, partnership)?:

Initiate a pilot implementation in a selected micro-scale biogas unit to validate real-time performance and operational impact.

Enhance the AI model using additional datasets to improve scalability across different climatic and feedstock conditions.

Establish partnerships with biogas plant manufacturers, clean energy startups, and local government bodies for deployment.

Seek research collaborations and apply for grants or incubation programs to support technical refinement and commercialization.

Optimize the cloud-based monitoring platform and





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Enhance the AI model using additional datasets to improve scalability across different climatic and feedstock conditions.

Establish partnerships with biogas plant manufacturers, clean energy startups, and local government bodies for deployment.

Seek research collaborations and apply for grants or incubation programs to support technical refinement and commercialization.

Optimize the cloud-based monitoring platform and mobile dashboard for large-scale rollout.

Develop a phased expansion roadmap targeting rural communities, agricultural sectors, and decentralized energy networks.

Outline your 3–5 year vision for the project.:

Deploy the AI-enabled biogas forecasting system across multiple rural and community-level anaerobic digestion units at regional and national scales. Develop a fully integrated smart waste-to-energy platform combining AI forecasting, real-time control automation, and mobile-based monitoring. Establish strategic partnerships with governments, renewable energy companies, and sustainability organizations for large-scale adoption. Transition toward commercialization through a scalable SaaS or licensing model for decentralized energy operators. Contribute to measurable carbon reduction goals and support global clean energy and circular economy initiatives.



How do you plan to ensure adaptability to future



sustainability organizations for large-scale adoption. Transition toward commercialization through a scalable SaaS or licensing model for decentralized energy operators. Contribute to measurable carbon reduction goals and support global clean energy and circular economy initiatives.

How do you plan to ensure adaptability to future technological or environmental changes?:

Design the system with a modular and upgradeable architecture so new AI models, sensors, or optimization techniques can be integrated easily. Enable continuous learning by retraining the model periodically with new regional and environmental data. Use cloud-based deployment to allow remote updates, security patches, and performance improvements. Incorporate scalable IoT standards to ensure compatibility with future sensor technologies. Monitor environmental trends and policy changes to adjust system parameters and sustainability targets accordingly. Maintain research collaborations to stay aligned with emerging advancements in AI, renewable energy, and waste management technologies.

Reflection & Learning





Reflection & Learning

What was your biggest challenge and key learning experience?:

Biggest Challenge:

Modeling the nonlinear and dynamic behavior of anaerobic digestion processes.

Managing multiple interdependent variables like temperature, pH, feedstock type, and loading rate. Achieving stable and accurate predictions with limited and noisy real-world data.

Key Learning Experience:

Combining domain knowledge with data-driven AI significantly improves model reliability.

Explainable AI is essential to build operator trust and ensure practical adoption.

Scalability, usability, and interpretability are as important as prediction accuracy in real-world systems.

How has this project shaped your understanding of sustainability and innovation?:

Sustainability Perspective:

Both projects focus on reducing reliance on fossil fuels—biogas replaces LPG or firewood, solar reduces electricity consumption, and waste heat recovery maximizes energy already available.

They emphasize circular economy principles: using organic waste for energy (biogas) or capturing heat that would otherwise be lost.

These systems promote energy efficiency at the community or household level, showing that sustainability isn't just about large-scale solutions—it can start small but smart.

Innovation Perspective:





How has this project shaped your understanding of sustainability and innovation?:

Sustainability Perspective:

Both projects focus on reducing reliance on fossil fuels—biogas replaces LPG or firewood, solar reduces electricity consumption, and waste heat recovery maximizes energy already available.

They emphasize circular economy principles: using organic waste for energy (biogas) or capturing heat that would otherwise be lost.

These systems promote energy efficiency at the community or household level, showing that sustainability isn't just about large-scale solutions—it can start small but smart.

Innovation Perspective:

Integrating solar PV with biogas shows creative hybridization—combining two technologies for a more reliable, efficient solution.

Using PCM (phase change materials) or heat exchangers in the waste heat recovery project shows innovative thinking in energy storage, not just production.

Both ideas demonstrate problem-driven innovation: they identify real issues (energy insecurity, waste management, inefficiency) and design practical, implementable solutions.

If you could restart, what would you do differently?:

Start with a smaller, modular prototype first

Incorporate more real-time monitoring

Focus on user experience earlier

Explore alternative materials and designs

Plan scalability and adaptability

What advice would you give to others addressing similar SDG problems?:





If you could restart, what would you do differently?:

- Start with a smaller, modular prototype first
- Incorporate more real-time monitoring
- Focus on user experience earlier
- Explore alternative materials and designs
- Plan scalability and adaptability

What advice would you give to others addressing similar SDG problems?:

- Start with the problem, not the tech
- Think in systems, not components
- Iterate rapidly
- Leverage local resources
- Prioritize simplicity and usability
- Measure impact scientifically
- Collaborate and share knowledge

Other Information

Provide the details about limitations or risks:

Data Dependency, Limited Dataset Size, Environmental Variability, Technical Infrastructure Requirements, Adoption & Training Challenges, Scalability Risks

Provide the details about compliance with Industry or Regulatory Standards:

Environmental Regulations, Renewable Energy Standards, Data Protection & Cybersecurity, IoT & Electrical Safety Standards, Quality & Operational Standards, Scalability & Reporting Compliance

Has this project been presented, published, or



Other Information

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Data Dependency, Limited Dataset Size, Environmental Variability, Technical Infrastructure Requirements, Adoption & Training Challenges, Scalability Risks

Provide the details about compliance with Industry or Regulatory Standards:

Environmental Regulations, Renewable Energy Standards, Data Protection & Cybersecurity, IoT & Electrical Safety Standards, Quality & Operational Standards, Scalability & Reporting Compliance

Has this project been presented, published, or awarded elsewhere?:

No

Have you participated in the past YESIST12 editions?:

No